SRI SAI SNIGDHA ANANTHARAJU

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PROFESSIONAL SUMMARY

I'm an urban data scientist who uses geospatial analysis, machine learning, and civic tech to tackle complex challenges in cities. At NYU CUSP, I've built predictive models for flood risk (90% accuracy), building energy performance (10K+ buildings), and spatial access gaps (2,000+ zones). I also bring a consulting mindset while translating messy urban data into actionable insights for planners, policymakers, and public agencies to drive inclusive, data-informed change.

EDUCATION

New York University, Tandon School of Engineering — *New York, NY*

Master of Science (MS), Applied Urban Science and Informatics [GPA: 3.945/4] — *May 2026*

- Recipient of CUSP Merit Scholarship of $18,000
- Winner, *Best Technical Contribution*, Global Data Dive at King's College London
- Recipient, CUSP Experiential Scholarship for guided research under Prof. Anton Rozhkov

School of Planning and Architecture, Vijayawada — *Vijayawada, India*

Bachelor of Planning, Urban and Regional Planning [GPA: 3.88/4] — *May 2024*

RELEVANT EXPERIENCE

GIS Consultant, Data Services, New York University — *May 2025-present*

- Deliver technical GIS support and spatial data consultation across research projects.
- Build custom ArcGIS workflows for network analysis, hotspot analysis, clustering and other spatial analyses.
- Present mapping techniques and analytical approaches in workshops for tech & non-tech academic audiences.

Graduate Assistant, Center for Urban Science + Progress, New York University — *November 2024 – present*

- Organized and categorized 100+ spatial datasets for reuse in climate, infrastructure, and mobility research.
- Coordinate logistics for events and programs, ensuring seamless execution while managing communications.

Data Science & Machine Learning Intern, School of Planning and Architecture, Bhopal — *June 2023 – July 2023*

- Extracted and analyzed 5,000+ accessibility-related data points using Python (BeautifulSoup, Selenium).
- Applied BERT, VADER, and GAN models to classify and evaluate heritage accessibility compliance.
- Supported the development of a Government of India accessibility mapping tool impacting 10,000+ users.

Urban Planning Intern, City Planning Department, Vijayawada Municipal Corporation — *June 2022 – August 2022*

- Conducted traffic flow analysis & proposed vending zones, NMT lanes leading to 12% reduction in traffic accidents.
- Contributed to the corridor improvement plan by integrating pedestrian and non-motorized transport-friendly features leading to around 25% increase in pedestrian safety and smoother traffic flow along 6+ critical corridors.
- Facilitated 4+ cross-sectoral stakeholder meetings to develop a TOD plan, impacting 25,000+ daily commuters.

SKILLS

- **Programming and GIS**: Python, ArcGIS Pro, QGIS, Google Earth Engine, EPANET, GeoDa
- **Visualization & Tools:** Tableau, PowerBI, Folium, MS Excel, Canva
- **Transportation & Planning Tools:** PTV Vissim, TOD Analysis, Accessibility Audits, Land Use & Zoning
- **Soft Skills & Communication:** Public Speaking, Research Synthesis, Technical Writing, Workshop Facilitation, Interdisciplinary Collaboration

RELEVANT PROJECTS

SoftWhere: Optimizing Housing Decisions to Minimize Commute Times — *May 2025*

- Designed a geospatial decision-support tool recommending housing locations based on subway commute efficiency using GTFS data and reverse-time Dijkstra algorithm.
- Modeled NYC's subway network as a time-dependent graph, integrating MTA schedules, hexagonal tessellation, and spatial proximity metrics.
- Developed commute accessibility scores for over 2,000 hexagons, visualized using interactive heatmaps to identify transit-rich yet underappreciated neighborhoods
- Proposed planning applications for housing equity, sustainable commuting, and transit-oriented location strategies.

Predicting Flood Vulnerability in Florida Using GIS and Machine Learning *May 2025*

- Integrated multi-source spatial datasets (FEMA flood zones, NOAA DEMs, NRI index, USGS 3DEP, FLDEP land use) for comprehensive risk mapping.
- Applied Hotspot and Network Service Area Analysis to identify emergency access gaps..
- Built classifiers using Forest-Based and SVM models to detect high-risk future zones.
- Evaluated exposure criteria including population density >1,000/sq mi, location of critical infrastructure in flood zones, and emergency access delays >10 minutes.

Invisible Boundaries: Network Analysis of Healthcare Access in Houston, TX *March 2025*

- Developed a network model in ArcGIS Pro to analyze healthcare accessibility in Houston, TX.
- Computed least-time travel paths to healthcare facilities, incorporating various speed limits and road hierarchy.
- Performed spatial inequity assessments using service area analyses, pinpointing underserved regions with >20-minute emergency response delays, highlighting demographic and spatial disparities.

New York: Skyline vs. Sustainability – Factors affecting NYC Buildings' Energy Efficiency *February 2024*

Won 'Best Technical Contribution' at the Global Data Dive (3-day Hackathon) held by King's College London

- Preprocessed building energy data (LL84) for 10000+ buildings in Manhattan, NYC
- Analyzed the impact of intrinsic and external factors on building energy efficiency in New York City.
- Assessed Local Law 97 compliance, identifying $218M in annual energy costs and associated penalties.
- Developed data-driven insights to help policymakers, building owners, and investors make decisions on energy.

Impact of Crime-Induced fear on Subway Ridership in Brooklyn, NY *December 2024*

- Analyzed MTA subway ridership data to investigate the spatial and temporal effects of a crime incident in Brooklyn.
- Mapped the spatial gradient of crime-induced ridership changes, identifying declines at stations along same line.
- Performed statistical validation with KS2 tests, detecting ridership reductions ($p < 0.05$) post-incident.
- Visualized commuter behavior, revealing a 25% decrease in afternoon ridership and a gradual recovery in 3 days.

Climate Change Mitigation Strategies for Chennai, India *December 2023*

Secured first place at a national level (India) in Eco-centric Strategies for mitigating Climate Change

- Developed a comprehensive analysis framework for assessing climate change-induced disasters in Chennai, focusing on flood risk mapping and drought prevention.
- Conducted flood risk mapping, vulnerability analysis, water resilience mapping for the Pallikaranai Marshland
- Utilized satellite imagery to evaluate the Urban Heat Island effect through LST and NDVI.
- Evaluated sea-level rise using Galdit's Model that projected a change of 0.18 feet in 100 years.
- Proposed eco-centric strategies like carbon sequestration, bioswales and a marshland regeneration project.

Vijayawada City-Wide Freight Transport Analysis and Emission Mitigation Strategies *May 2023*

- Mapped spatial distribution of pollutants using ArcGIS, pinpointing high-emission zones.
- Optimized logistics by evaluating trip patterns of 65 daily freight vehicles, with medium commercial vehicles averaging 1,125 km per trip, and proposed solutions to reduce congestion and improve routing efficiency.
- Analyzed freight operations across Vijayawada, calculating vehicle emissions of 9,118 kg/day, including PM_{10}, $PM_{2.5}$, CO, NO_x, and SO_2, and developed phased EV adoption plans to achieve a 62% reduction in emissions
- Designed a phased electrification strategy for freight vehicles, projecting emission reductions of 1,819.75 kg/day in Phase 1 and 5,726.86 kg/day in Phase 2.

Detailed Development Plan – Mudichur and Vandalur, Tamil Nadu *December 2022*

- Designed a sustainable water supply system using EPANET to perform hydraulic modeling and pressure optimization, reducing system water loss and improving efficiency by 20%.
- Modeled 24/7 potable water distribution for 10,000+ residents, reducing dependency on underground water and supporting UN SDGs 6 (Clean Water) and 11 (Sustainable Cities).
- Performed system-wide water demand analysis and identified critical zones for pressure improvements.
- Coordinated land-use planning with first/last mile mobility interventions, targeting a 15% reduction in projected vehicle-kilometers traveled (VKT) and enhanced access equity.

PUBLICATION

- *Book chapter:* Saha K., Snigdha, A. Khare R. GIS to Machine Learning- Application of Emerging Tools to assess accessible built environment in Indian cities. In Yadav, V., Mishra, M., Gavsker, K. (Eds.) The Built Environment and Public Health in India- Opportunities, Challenges, and Future Pathways. Springer Nature. (accepted)